

Equations with Fractions (Homework)

Name
Date
Course

Solve.

Answers

$$\textcircled{1} 5 + \frac{x}{3} = 4x + \frac{7}{9}$$

$$x = \frac{38}{33}$$

$$\textcircled{2} \frac{1}{6} - y = \frac{3y}{4} + \frac{1}{12}$$

$$y = \frac{1}{21}$$

$$\textcircled{3} -\frac{1}{8} + \frac{b}{6} = 2 + b$$

$$b = -\frac{51}{20}$$

$$\textcircled{4} \frac{3x}{2} + \frac{1}{4} = \frac{5x}{8} + \frac{x}{4}$$

$$x = -\frac{2}{5}$$

$$\textcircled{5} 2 - \frac{1}{9y} = \frac{2}{y} + \frac{5}{9}$$

$$y = \frac{19}{13}$$

$$\textcircled{6} \frac{7}{2b} = \frac{4}{b} + \frac{1}{2b^2}$$

$$b = -1$$

$$\textcircled{7} \frac{1}{10} + \frac{1}{x} = \frac{3}{5x} - \frac{4}{5}$$

$$x = -\frac{4}{9}$$

$$\textcircled{8} \frac{7}{8y^3} - \frac{1}{4y^2} = \frac{1}{2y^2}$$

$$y = \frac{7}{6}$$

$$\textcircled{9} b + \frac{16}{b} = 10$$

$$b = 8 \text{ or } 2$$

$$\textcircled{10} 1 + \frac{3}{4x} = -\frac{x}{4}$$

$$x = -1 \text{ or } -3$$

$$\textcircled{11} y^2 + \frac{11y}{2} = 3$$

$$y = \frac{1}{2} \text{ or } -6$$

$$\textcircled{12} \frac{3}{2} - \frac{4}{b} = -\frac{2}{b^2}$$

$$b = 2 \text{ or } \frac{2}{3}$$

$$\textcircled{13} \frac{1}{x} + \frac{9}{x^2} = -\frac{8}{x^3}$$

$$x = -8 \text{ or } -1$$

$$\textcircled{14} \frac{6}{y^2} = \frac{7}{y^3} + \frac{20}{y^4}$$

$$y = -\frac{4}{3} \text{ or } \frac{5}{2}$$

$$\textcircled{15} \frac{2}{b-6} - \frac{4b}{b^2-9b+18} = \frac{8}{b-3}$$

$$b = \frac{21}{5}$$

$$\textcircled{16} \frac{7}{x+2} + \frac{3}{x-2} = \frac{1}{x^2-4}$$

$$x = \frac{9}{10}$$

$$\textcircled{17} \frac{5}{4y^2-12y} = \frac{9}{4y} - \frac{2}{y-3}$$

$$y = 32$$

$$(18) \frac{-3}{b^2+b-2} + \frac{2}{b-1} = \frac{1}{b+2}$$

No Solution

$$(19) -\frac{10x}{25x^2-36} + \frac{3}{5x-6} = \frac{5}{5x+6}$$

$$x = \frac{12}{5}$$

$$(20) \frac{y}{5y+2} + \frac{1}{y+1} = \frac{-6}{5y^2+7y+2}$$

$$y = -2 \text{ or } -4$$

$$(21) \frac{-4}{b-5} + \frac{2}{b+3} = \frac{b^2-57}{b^2-2b-15}$$

$$b \neq 5$$

$$b = -7$$

$$(22) \frac{x^2}{x^2-7x+12} + \frac{5}{x-4} = \frac{3}{x-3}$$

$$x = 1 \text{ or } -3$$

$$(23) \frac{4}{y-1} - \frac{4}{y+1} - \frac{7}{y^2-1} = \frac{y^2}{y^2-1}$$

No Solution

Both solutions are extraneous

$$(24) \frac{2}{b-5} = \frac{b^2+3}{3b^2-15b} - \frac{5}{3b}$$

$$b = 7 \text{ or } 4$$

$$(25) \frac{2x}{x-1} + \frac{6}{x+7} = \frac{25x-3}{x^2+6x-7}$$

$$x = 3 \text{ or } -\frac{1}{2}$$

* You must use FOIL.

$$(26) \frac{x+2}{3x-2} + \frac{5}{x-8} = \frac{x+7}{3x^2-26x+16}$$

$$x = 3 \text{ or } -11$$

* Be very careful with signs. Distribute properly.

$$(27) \frac{8}{y-2} = -\frac{5y-4}{y-1} + \frac{20y-5}{y^2-3y+2}$$

$$y = 5 \text{ or } \frac{1}{5}$$

* Watch your signs. Distribute properly.